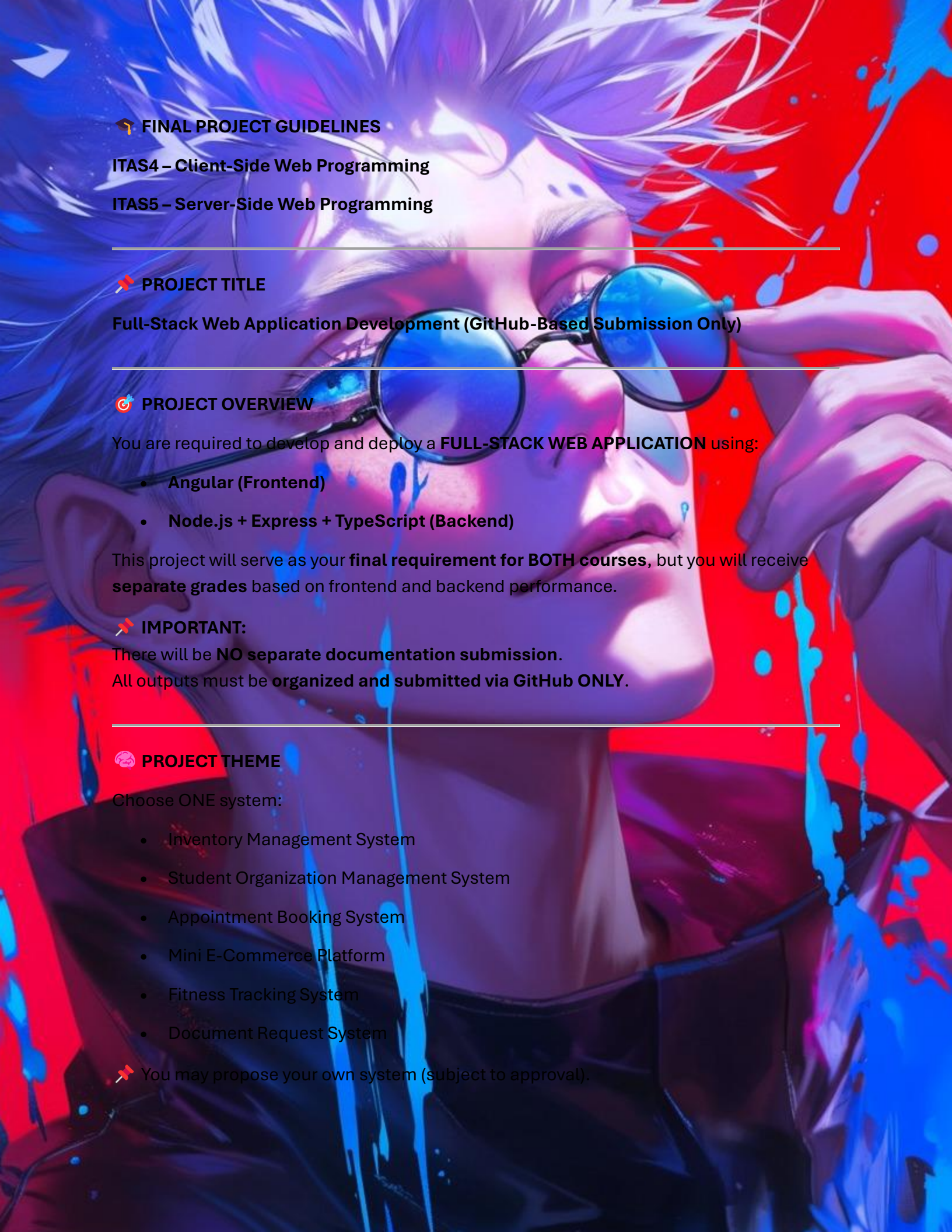




📌 FINAL PROJECT GUIDELINES

ITAS4 – Client-Side Web Programming

ITAS5 – Server-Side Web Programming

🚩 PROJECT TITLE

Full-Stack Web Application Development (GitHub-Based Submission Only)

🎯 PROJECT OVERVIEW

You are required to develop and deploy a **FULL-STACK WEB APPLICATION** using:

- Angular (Frontend)
- Node.js + Express + TypeScript (Backend)

This project will serve as your **final requirement for BOTH courses**, but you will receive **separate grades** based on frontend and backend performance.

🚩 IMPORTANT:

There will be **NO separate documentation submission**.

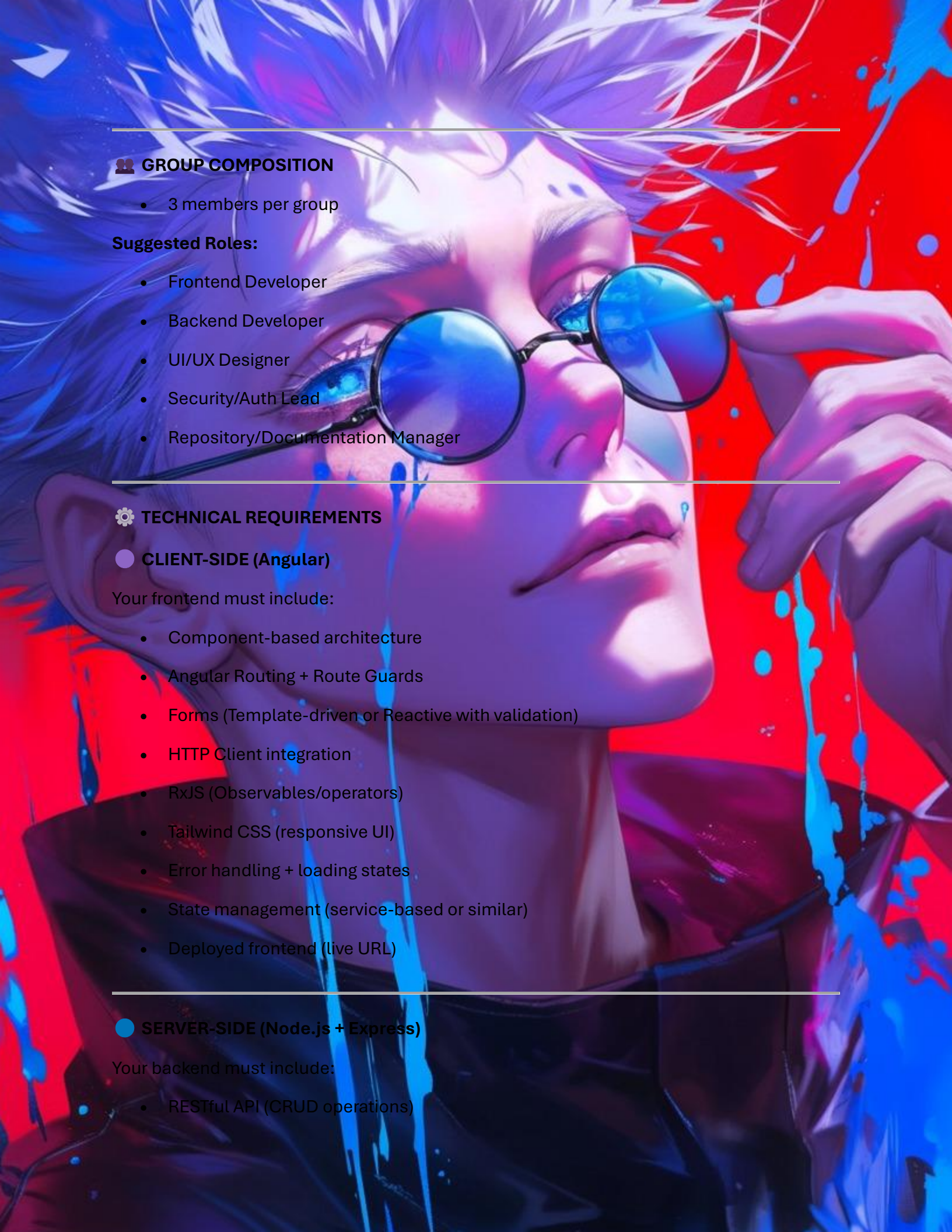
All outputs must be **organized and submitted via GitHub ONLY**.

🧠 PROJECT THEME

Choose ONE system:

- Inventory Management System
- Student Organization Management System
- Appointment Booking System
- Mini E-Commerce Platform
- Fitness Tracking System
- Document Request System

🚩 You may propose your own system (subject to approval).



GROUP COMPOSITION

- 3 members per group

Suggested Roles:

- Frontend Developer
 - Backend Developer
 - UI/UX Designer
 - Security/Auth Lead
 - Repository/Documentation Manager
-

TECHNICAL REQUIREMENTS

CLIENT-SIDE (Angular)

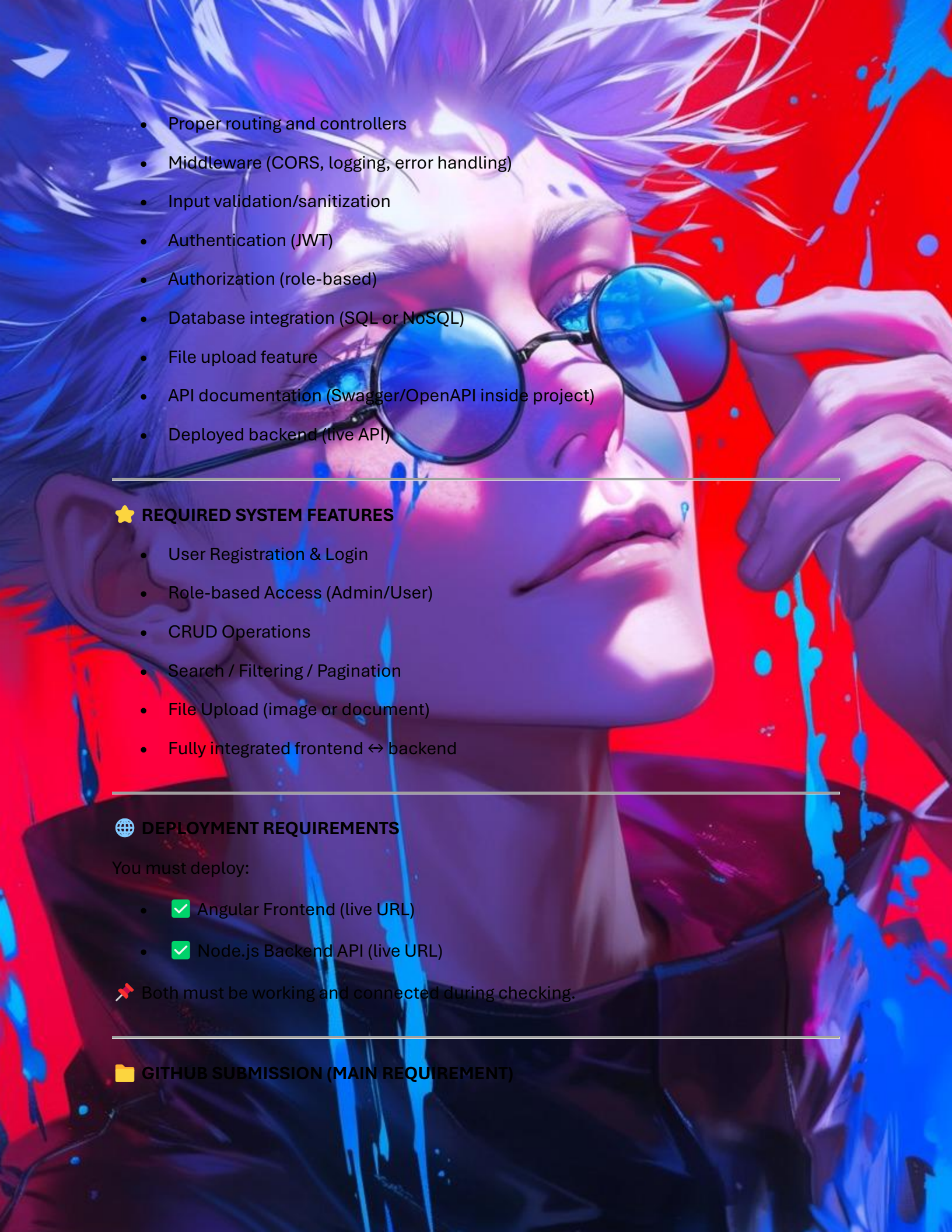
Your frontend must include:

- Component-based architecture
 - Angular Routing + Route Guards
 - Forms (Template-driven or Reactive with validation)
 - HTTP Client integration
 - RxJS (Observables/operators)
 - Tailwind CSS (responsive UI)
 - Error handling + loading states
 - State management (service-based or similar)
 - Deployed frontend (live URL)
-

SERVER-SIDE (Node.js + Express)

Your backend must include:

- RESTful API (CRUD operations)

- 
- Proper routing and controllers
 - Middleware (CORS, logging, error handling)
 - Input validation/sanitization
 - Authentication (JWT)
 - Authorization (role-based)
 - Database integration (SQL or NoSQL)
 - File upload feature
 - API documentation (Swagger/OpenAPI inside project)
 - Deployed backend (live API)
-

★ REQUIRED SYSTEM FEATURES

- User Registration & Login
 - Role-based Access (Admin/User)
 - CRUD Operations
 - Search / Filtering / Pagination
 - File Upload (image or document)
 - Fully integrated frontend ↔ backend
-

🌐 DEPLOYMENT REQUIREMENTS

You must deploy:

- ☒ Angular Frontend (live URL)
 - ☒ Node.js Backend API (live URL)
- 📌 Both must be working and connected during checking.
-

📁 GITHUB SUBMISSION (MAIN REQUIREMENT)

All groups must submit **ONE GitHub repository ONLY.**

👉 Use: [Github](#)

📁 REQUIRED REPOSITORY STRUCTURE

project-name/

```
|
|
|— client/    → Angular Application
|— server/    → Node.js API
|
|— README.md  → MAIN PROJECT GUIDE
|— .env       → Environment variables
|— screenshots/ → UI screenshots and API Testing
```

📄 REQUIRED README CONTENT

Your **README.md** must include:

1. Project Overview

- System name and description

2. Live Links

- Frontend URL
- Backend API URL

3. Tech Stack

- Frontend: Angular + Tailwind
- Backend: Node.js + Express + TypeScript
- Database used

4. Setup Instructions

- How to run frontend
- How to run backend

5. API Overview

- List of main endpoints

6. Features Implemented

- Clearly list all features

7. Screenshots

- UI screenshots
- API testing (Postman or similar)

✖ No separate PDF allowed. Everything must be inside GitHub.

GRADING CRITERIA

● CLIENT-SIDE (50%)

- UI/UX Design (Tailwind, responsiveness) – 15%
- Angular Architecture – 15%
- API Integration & RxJS – 10%
- Forms & Validation – 5%
- Deployment – 5%

● SERVER-SIDE (50%)

- REST API Design – 15%
- Database Integration – 10%
- Authentication & Security – 10%
- Middleware & Error Handling – 5%
- Deployment – 5%
- README/API Documentation – 5%



🖌️ FINAL PRESENTATION

Each group must:

- Present their **LIVE system**
- Show GitHub repository
- Explain system architecture
- Demonstrate features
- Answer technical questions

⚠️ IMPORTANT REMINDERS

- ❌ No GitHub = No Project
- ❌ No Deployment = Deduction
- ❌ No README = Incomplete Project

🚀 FINAL NOTE

This project represents a **real-world full-stack system**.

Focus on:

- Clean code
- Proper structure
- Security
- Working deployment

👉 Treat your GitHub repository as your **portfolio project**.

DEADLINE: 2nd Week of May 2026.

